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Candidate Number							

HONG KONG DIPLOMA OF SECONDARY EDUCATION EXAMINATION BYTOPIC

MATHEMATICS Extended Part

Module 2 (Algebra and Calculus)

Question-Answer Book

Time allowed : $2\frac{1}{2}$ hours

This paper must be answered in English

INSTRUCTIONS

(1) After the announcement of the start of the examination, you should first write your Candidate Number in the space provided on Page 1 and stick barcode labels in the spaces provided on Pages 1.

(2) This paper consists of TWO sections, A and B.

(3) This paper carried 100 marks.

(4) Attempt ALL questions in this paper. Write your answers in the spaces provided in this Question-Answer Book. Do not write in the margins. Answers written in the margins will not be marked.

(5) Graph paper and supplementary answer sheets will be supplied on request. Write your Candidate Number, mark the question number box and stick a barcode label on each sheet, and fasten them with string INSIDE this book.

(6) Unless otherwise specified, all working must be clearly shown.

(7) Unless otherwise specified, numerical answers must be exact.

(8) The diagrams in this paper are not necessarily drawn to scale.

(9) No extra time will be given to candidates for sticking on the barcode labels or filling in the question number boxes after the 'Time is up' announcement.

FORMULAS FOR REFERENCE

$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$	$\sin A + \sin B = 2 \sin \frac{A+B}{2} \cos \frac{A-B}{2}$
$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$	$\sin A - \sin B = 2 \cos \frac{A+B}{2} \sin \frac{A-B}{2}$
$\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$	$\cos A + \cos B = 2 \cos \frac{A+B}{2} \cos \frac{A-B}{2}$
$2 \sin A \cos B = \sin(A+B) + \sin(A-B)$	$\cos A - \cos B = -2 \sin \frac{A+B}{2} \sin \frac{A-B}{2}$
$2 \cos A \cos B = \cos(A+B) + \cos(A-B)$	
$2 \sin A \sin B = \cos(A-B) - \cos(A+B)$	

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SECTION A (74 marks)

1. In the expansion of $(1+ax)^n$, the coefficient of x^{n-2} is 54 and the coefficient of x is -12.

   (a) Find a and n. (4 marks)

   (b) Find the coefficient of x^3. (2 marks)

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2. In the expansion of $(2+4x)^n$, where n is a positive integer, let u_k be the coefficient of x^k. It is given that $2u_1 + 8u_0 = u_2$.

(a) Find n. (3 marks)

(b) Hence, evaluate $0 \cdot u_0 + 1 \cdot u_1 + 2 \cdot u_2 + \cdots + n \cdot u_n$. (2 marks)

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3. Consider the expansion of $\left(1+\frac{1}{x}\right)^2\left(kx+\frac{1}{x}\right)^4$, where k is a constant. It is given that the constant term in the expansion is 0.

(a) Find the value(s) of k. (3 marks)

(b) Someone claims that for any positive integer n, the coefficient of x^{-n} in the expansion of $\left(1+\frac{1}{x}\right)^2\left(kx+\frac{1}{x}\right)^n$ must not be 1. Do you agree? Explain your answer. (3 marks)

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4. In the expansion of $(au+t)^n$, where a, t are non-negative integers and n is a positive integer, write $(au+t)^n = \sum_{r=0}^{n} \alpha_r u^r$. It is given that $\alpha_1 = 12$, $\alpha_2 = 60$, and $\alpha_3 = 160$.

(a) Find a, t and n. (4 marks)

(b) Write $(au+bv)^n = \sum_{r=0}^{n} \beta_r v^r$, where b is an integer. Prove that $\beta_4 - \alpha_4$ is divisible by b^2u+2. (3 marks)

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5. Consider the expansion of $(x^{-2}+x)^5(x^3-x^{-2})^2$.

(a) Find the coefficient of x^{-4} in the expansion. (3 marks)

(b) Someone claims that the coefficient of x^2 in the expansion is 0. Do you agree? Explain your answer. (3 marks)

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6. Consider the expansion of $(1+kx)^n$, where n is an integer greater than 2, and k is a non-zero constant.

It is given that the sum of the coefficient of x and the coefficient of x^2 is 50. In addition, the ratio of the coefficient of x^2 to the coefficient of x^3 is $1:2$.

(a) Find the values of n and k. (4 marks)

(b) Let $(1+kx)^n = P(x)$. If a constant term exists in the expansion of $\left(x^2 - \dfrac{1}{2x}\right)^6 P(x)$, find the value of this constant term. (4 marks)

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7. Let n and k be positive integers with $n > 2$. Consider the expressions $f(x) = (1 + kx)^n$ and $g(x) = (kx - 2)(1 + x)^n$.

(a) (i) Express the coefficient of x and the coefficient of x^2 in the expansion of $f(x)$ in terms of n and k. (2 marks)

(ii) It is given that the coefficient of x^2 in the expansion of $f(x)$ is 12 times the coefficient of x. Prove that $k(n - 1) = 24$. (2 marks)

(b) (i) Express the coefficient of x in the expansion of $g(x)$ in terms of n and k. (2 marks)

(ii) It is given that the sum of the constant term and the coefficient of x in the expansion of $g(x)$ is -6. Find the values of n and k. (3 marks)

(c) Using the values of n and k obtained in (a) and (b), find the coefficient of x^2 in the expansion of $g(x)$. (3 marks)

8. Let n be an integer greater than 3, and let a and b be non-zero constants. Consider the following two expressions:

$$f(x) = \left(ax^2 - \frac{b}{x}\right)^3 \left(ax^2 + \frac{2b}{x}\right)^n$$

$$g(x) = \left(ax^2 + \frac{b}{x}\right) \left(ax^2 - \frac{kb}{x}\right)^{n+2}$$

where k is a non-zero constant.

(a) (i) Prove that the constant term (i.e. the coefficient of x^0) in the expansion of $f(x)$ is $2n(n-1)a^n b^5$. (3 marks)

(ii) Find the coefficient of x^3 in the expansion of $f(x)$ in terms of a, b, and n. (2 marks)

(b) It is given that the constant term in the expansion of $g(x)$ is also $2n(n-1)a^n b^5$.

(i) Express k in terms of n. (2 marks)

(ii) Hence, prove that the coefficient of x^3 in the expansion of $g(x)$ can be written as

$$\frac{(n+2)(n-1)(n-2)}{n} \cdot a^{n+1} b^4.$$

(2 marks)

(c) Now compare the coefficients of x^3 in the two functions. Given that $a = 1$ and $b = 1$, if the coefficient of x^3 in $f(x)$ is equal to the coefficient of x^3 in $g(x)$, find the values of n and k. (3 marks)

9. A sequence of functions $F_n(x)$ is defined by

$$F_n(x) = \sum_{k=1}^{n} \left[\binom{n}{k} \cdot \left(\frac{x}{1+2x} \right)^k \right].$$

(a) (i) By considering the expansion of the binomial theorem, prove that

$$1 + F_n(x) = \left(\frac{1+3x}{1+2x} \right)^n.$$

(2 marks)

(ii) Consider $F_n(x) = c_0 + c_1 x + c_2 x^2 + \cdots + c_n x^n$. Hence, find c_1 in terms of n.

(2 marks)

(b) Consider another function $G_n(x) = (1+2x)^n \cdot [1 + F_n(x)]$.

(i) Prove that $G_n(x) = (1+3x)^n$. (1 mark)

(ii) Let $G_n(x) = b_0 + b_1 x + b_2 x^2 + \cdots + b_n x^n$. Prove that for all $2 \le r \le n$,

$$\frac{b_r}{b_{r-1}} = \frac{3(n-r+1)}{r}.$$

(4 marks)

(c) It is given that the ratio of three consecutive coefficients in the expansion of $G_n(x)$ is $b_{r-1} : b_r : b_{r+1} = 1 : 3 : 7$. Find the values of n and r. (3 marks)

SECTION B (26 marks)

10. It is given that matrix $M = \begin{pmatrix} 2 & -1 \\ 3 & -2 \end{pmatrix}$ and $I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$.

(a) Prove that $M^2 = I$. (2 marks)

(b) Consider the matrix series $X = \sum_{r=0}^{5} \binom{5}{r} 2^{5-r} M^r$.

By using the binomial theorem and the result of (a), express the matrix X in the form $aI + bM$ (where a and b are constants), and find the exact value of the matrix X. (5 marks)

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11. It is given that $A = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$ and $I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$.

(a) Prove by mathematical induction that for all positive integers n,

$$A^n = \frac{1}{2}\begin{pmatrix} 3^n + (-1)^n & 3^n - (-1)^n \\ 3^n - (-1)^n & 3^n + (-1)^n \end{pmatrix}.$$

(5 marks)

(b) Consider a matrix B_n defined by

$$B_n = \sum_{r=0}^{n} \binom{n}{r} A^{2r}$$

where n is a positive integer.

(i) Prove that $A^2 - 2A - 3I = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$. (1 mark)

(ii) Without using comparison of coefficients, by expanding $(I + A^2)^n$ using the binomial theorem, prove that

$$B_n = \frac{1}{2}\begin{pmatrix} 10^n + 2^n & 10^n - 2^n \\ 10^n - 2^n & 10^n + 2^n \end{pmatrix}.$$

(4 marks)

(c) In the Cartesian plane, the three vertices of $\triangle OPQ$ are the origin $O(0,0)$, $P(1,0)$, and $Q(0,1)$. Two points P' and Q' satisfy the matrix equations:

$$\begin{pmatrix} x_{P'} \\ y_{P'} \end{pmatrix} = B_n \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} x_{Q'} \\ y_{Q'} \end{pmatrix} = B_n \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

where B_n is the matrix defined in (b)(ii).

(i) Using the result of (b)(ii), find the determinant of B_n, $\det(B_n)$, and express your answer as a single exponential expression. (2 marks)

(ii) It is given that the area of $\triangle OP'Q'$ can be calculated by Area $= \frac{1}{2}|x_{P'}y_{Q'} - x_{Q'}y_{P'}|$. Prove that the area of $\triangle OP'Q'$ is 20^n times the area of $\triangle OPQ$.

(2 marks)

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12. It is given that n is a positive even integer. In a 2-dimensional rectangular coordinate plane, two vectors $\mathbf{u}_n$ and $\mathbf{v}_n$ depending on n are defined by

$$\mathbf{u}_n = a_n\mathbf{i} + 2^n\mathbf{j} \quad \text{and} \quad \mathbf{v}_n = b_n\mathbf{i} - \mathbf{j}$$

where the coordinate components are defined by the following binomial series:

$$a_n = \sum_{r=0}^{n} \binom{n}{r} (\sqrt{3})^r \quad \text{and} \quad b_n = \sum_{r=0}^{n} \binom{n}{r} (-\sqrt{3})^r.$$

By using the binomial theorem, prove that the dot product of these two vectors is a constant identically equal to zero (i.e. $\mathbf{u}_n \cdot \mathbf{v}_n = 0$), and describe the geometric relationship between $\mathbf{u}_n$ and $\mathbf{v}_n$. (5 marks)

END OF PAPER